GROCERY

sim ABM

Agent-Based Model for Consumer Behaviour

& Supply Chain Stress-Testing

USER MANUAL v2.0

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GROCERYsim is an open agent-based model (ABM) for stress-testing the resilience of grocery supply chains under crisis scenarios. It simulates a retail environment as autonomous agents: consumers with individual cognitive traits, shelves with finite inventory, and logistics with realistic lead times.

From their interactions, system-level behaviour emerges: resilience, fragility, and adaptation under shock. The model is calibrated to real participant survey data and reproduces panic buying, substitution cascades, and logistics disruption.

CITATION
Duric, Ivan (2026). GROCERYsim Agent-Based Model for Consumer Behaviour and Supply Chain Stress-Testing. IAMO XR Lab, SecureFood project, Horizon Europe Grant 101136583.

Target Users

- Researchers - calibrate scenarios, run sensitivity analyses, publish results.
- Policy makers - explore intervention strategies before crises occur.
- Retailers - validate contingency plans and test restocking policies.
- Students - understand complex adaptive systems in food supply chains.

2. Getting Started

When the app loads, click "Explore case studies" then "Launch simulation" on the Finland case to enter the main dashboard.

> The app ships with a built-in dataset. You can run a full simulation immediately without uploading any files.

The main dashboard has a left sidebar for simulation parameters and a tabbed main area for results: Data & Population, Interactive Demo, Scientific Analysis, Food Waste, Per-Product, Policy Analysis, Stakeholder View, Sensitivity Analysis, Behavioural Theory, and Export.

Shows the status of the loaded dataset and allows uploading custom data to rebuild the agent population pool.

Status Messages

- Green - data loaded. Shows number of real participants, synthetic agents, and products.
- Yellow - bundled data not fully configured. Follow the instructions shown.
- Blue - no data loaded yet. Upload files and click Process Data.

Uploading Custom Data

Expand "Reload / Override Data Files" and upload:

- Firebase Export JSON - raw survey/behavioural data from Firebase Realtime Database.
- Product Catalogue JSON - master_products.json listing all SKUs, categories, and prices.

Set Population Pool Size (default 2000) and Number of Behavioural Archetypes (default 4), then click "Process Uploaded Files". Processing takes 5-30 seconds.

> Archetype proportions reflect real survey data. Typical: 37% Price Champion, 25% Green Buyer, 14% Health Optimizer, 24% Habitual Buyer.

All parameters are set in the left sidebar. Changes take effect on the next simulation run.

Parameter	Default	Description
Duration (Days)	60	Total simulation length. 30-90 days recommended.
Start Month	1 (Jan)	Sets seasonal demand multipliers for the run.
Base Daily Consumers	100	Mean shoppers per day. Auto-calibrated to store size.
Store Size	Small	Neighbourhood (<200/day), Supermarket (200-500), Hypermarket (500+).

Logistics

Reorder Point (%)	30%	Stock level that triggers a new order.
Restock Target (%)	90%	Stock level to refill to when order arrives.
Lead Time (Days)	2	Days between placing and receiving an order.
Delivery Days	Mon/Thu	Days of the week when deliveries arrive.
Max Storage (%)	120%	Maximum stock above normal shelf capacity.

Crisis Scenario

Crisis Type	Supply Shock	Supply Shock, Demand Spike, Logistics Failure, or Combined.
Crisis Severity	0.5	0 = no impact, 1.0 = maximum disruption.
Crisis Start Day	15	Simulation day on which the crisis begins.
Crisis Duration	14 days	How many days the crisis condition persists.
Panic Multiplier	1.8	Demand amplification factor during crisis days.

Consumer Behaviour

Hoarding Threshold	0.3	Stock-out rate above which consumers start hoarding.
Substitution Rate	0.6	Probability a consumer accepts a substitute product.
Price Sensitivity	0.5	Mean sensitivity to price changes across the population.
Online Channel Share	0.15	Fraction of purchases diverted to online/delivery.

Runs a single two-scenario simulation (Baseline vs Crisis) with animated, real-time charts updating day by day.

Running a Simulation

- Set parameters in the sidebar.
- Click the orange "Run Simulation" button.
- Charts update live. Typical runtime: 5-20 seconds for 60 days.
- Once complete, all result tabs become active.

Result Tabs

- Financials - daily revenue, cumulative revenue, stock value, cost breakdown.
- Footfall - daily visitor counts, basket size, and conversion rate.
- Logistics - order events, delivery schedule, stockout frequency.
- Crisis Breakdown - side-by-side crisis vs baseline divergence charts.
- Behavioural Drift - preference shift heatmaps and archetype trajectory.
- By Buyer Type - per-archetype scorecards, bar charts, and transition table.

Key Metrics

- Revenue Gap - difference between Crisis and Baseline cumulative revenue.
- Stockout Rate - fraction of trips without desired item. >15% is critical.
- Fulfillment Rate - share of intended purchases completed. Target >85%.
- Panic Level - mean panic score across agents. Rises in week 2-3 of a crisis.
- FIES Score - Food Insecurity Experience Scale proxy. Higher = more food insecurity.

Runs a full Monte Carlo experiment with multiple stochastic replications of both scenarios to produce statistically robust results with confidence intervals.

3-Stage Workflow

- Stage 1: Set number of runs (5-50 recommended) and click "Run Monte Carlo". Runtime: 1-10 minutes.
- Stage 2: Review AI-generated optimisation recommendations to minimise stockouts while controlling waste.
- Stage 3: Explore results with confidence bands, violin plots, correlation heatmaps, and sensitivity tornado charts.

> Parameters with total Sobol index > 0.1 are influential. Focus policy interventions on these first.

7. Simulation Results Tabs

Financials

- Daily Revenue (Baseline vs Crisis) - day-by-day divergence.
- Cumulative Revenue - running total; gap widens as crisis persists.
- Average Basket Value - mean spend per shopper per day.
- Net Margin - revenue minus restocking, waste, and operational costs.

Footfall

- Daily Visitors - actual vs expected shopper count.
- Conversion Rate - percentage of visitors who complete a purchase.
- Queue Time - proxy for checkout congestion during peak crisis.

Logistics

- Order Events - bar chart of reorder triggers by day.
- Delivery Calendar - heatmap of delivery days and volumes.
- Stockout Map - which products stocked out and on which days.

Crisis Breakdown

Side-by-side comparison across all key metrics. Red = deterioration; green = improvement vs baseline.

Behavioural Drift

Consumer preferences captured daily per archetype. Shows which categories gained or lost share during crisis.

By Buyer Type

All results broken down by the four archetypes. Scorecards show baseline value and crisis delta. Transition table identifies the archetype with the largest behavioural shift.

Food Waste Dashboard

- Waste by Category - product groups generating the most waste by value and volume.
- Waste by Reason - expiry, overstock, damage, returns.
- Waste Trend - daily waste over simulation, Baseline vs Crisis.
- Waste Reduction Opportunities - AI-flagged products with disproportionate waste.

> Crisis scenarios typically reduce waste initially (stockouts clear stock) then spike post-crisis as over-ordering occurs during recovery.

Per-Product Deep-Dive

Select any SKU from the dropdown for a complete single-product analysis.

- Stock Chart - shelf stock level each day, with reorder and delivery events marked.
- Sales vs Demand - fulfilled vs attempted purchases. Gap = lost sales.
- Substitution Flow - products consumers switched to when this item stocked out.
- Price Elasticity - estimated demand response to price changes.

9. Policy Analysis

Test specific policy interventions by running a third scenario alongside Baseline and Crisis.

Available Policies

- Fat Tax - price surcharge on high-sugar/fat products.
- Healthy Subsidy - reduces price of nutritious staples.
- Supply Shock Buffer - pre-positions emergency stock before the crisis window.
- Purchase Limit - caps quantity any shopper can buy per item per visit.
- Labelling Intervention - increases green buyer preference for labelled products.

Reading Policy Results

- Economic panel - revenue and cost impact of the policy.
- Environmental panel - waste, carbon proxy, and food miles change.
- Welfare panel - FIES score and fulfillment rate across archetypes.
- Policy Effectiveness Score - composite index (0-100) from the optimiser.

Behavioural Theory Tab

- Theory of Planned Behaviour - attitude, subjective norm, and perceived behavioural control.
- Protection Motivation Theory - threat appraisal and coping appraisal under crisis.
- Dual-Process Model - System 1 (habitual) vs System 2 (deliberative) purchase decisions.
- Food Insecurity Experience Scale (FIES) - WHO-aligned food security measurement.

Export Tab

- Full Results CSV - all daily records for Baseline and Crisis scenarios.
- Stock Data CSV - product-level daily stock levels.
- Logistics Log CSV - all order, delivery, and stockout events.
- Waste Data CSV - itemised waste events with reason codes.
- PDF Report - auto-generated executive summary with key charts.

Four behavioural archetypes identified via K-Means clustering of real participant survey data.

Price Champion (37%)

Maximises value for money. High price sensitivity, low brand loyalty. First to switch to cheaper substitutes. Most responsive to subsidy and fat tax policies. Theory: Rational Choice, Price-Quality heuristic.

Green Buyer (25%)

Prioritises sustainability, organic, and locally sourced products. Willing to pay a premium. Low panic hoarding. Most affected by labelling interventions. Theory: Value-Belief-Norm, Pro-Environmental Behaviour.

Health Optimizer (14%)

Focuses on nutritional content, avoids processed foods. Moderate price sensitivity. Prefers familiar healthy brands. Theory: Health Belief Model, Self-Determination Theory.

Habitual Buyer (24%)

Follows routine shopping patterns. High brand loyalty, low substitution. Most resistant to change but most disrupted when preferred items stock out. Theory: Habit Theory, Theory of Planned Behaviour.

CITATION

Duric, Ivan (2026). GROCERYsim Agent-Based Model for Consumer Behaviour and Supply Chain Stress-Testing. IAMO XR Lab, SecureFood project, Horizon Europe Grant 101136583.

App shows "No data loaded"

Upload both JSON files in the Data & Population tab, or configure Streamlit Secrets.

"Process Uploaded Files" fails

Check you are uploading the correct files: Firebase export JSON and master_products.json. Both must be valid JSON under 200MB.

Charts are empty after simulation

Increase population pool size or crisis severity. Minimum pool size is 50.

"By Buyer Type" graphs show all zeros

Re-run the simulation. This issue occurred in older versions where agent metrics were captured after scheduler removal.

PDF report download fails

Run: pip install fpdf2. If the issue persists, use the CSV export instead.

App is slow on Scientific Analysis

Reduce Monte Carlo run count or use a shorter duration (30 days) for exploratory analysis.

Logos not visible on landing page

Ensure PNG files are in the static/ folder and enableStaticServing = true is in .streamlit/config.toml.

How to Cite

CITATION

Duric, Ivan (2026). GROCERYsim Agent-Based Model for Consumer Behaviour and Supply Chain Stress-Testing. IAMO XR Lab, SecureFood project, Horizon Europe Grant 101136583.

Contact & Support

For technical support or bug reports, contact the IAMO XR Lab via the SecureFood project portal at securefood-project.eu. Source code and documentation available on the project GitHub repository.